Neural Coding, Homework 4

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In [108…

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

1. Bernoulli Neuron: Fisher information and estimator

Assume a Bernoulli neuron with arbitrary tuning curve $f(s)$, i.e.

$$p(r|f(s)) = f(s)^r(1-f(s))^{1-r}, \qquad r \in \{0;1\}$$

Note: $r$ is discrete and $s$ is continuous.

(a) Compute the Fisher information $J_s$. *(2 points)*

We use the result from the lecture on how $J_s$ depends on the Fisher information J_f of the Bernoulli parameter f

$$J_s = \left[\frac{\partial f(s)}{\partial s}\right]^2 \cdot J_f$$

Where

$$\begin{aligned} J_f &:= Var\left(\frac{\partial}{\partial s}\log p\left(r|f(s)\right)\right) \\ &= -E_r\left[\left(\frac{\partial^2}{\partial f^2}\log p(r|f)\right)|f\right] \text{ (c.f. lecture)} \\ &= -E_r\left[\left(\frac{\partial^2}{\partial f^2}\log\left(f^r(1-f)^{1-r}\right)\right)|f\right] \\ &= -E_r\left[\frac{\partial^2}{\partial f^2}(r\log f + (1-r)\log(1-f))\,|f\right] \\ &= -E_r\left[\frac{\partial}{\partial f}\left(\frac{r}{f}-\frac{1-r}{1-f}\right)|f\right] \\ &= -E_r\left[-\frac{r}{f^2}-\frac{1-r}{(1-f)^2}|f\right] \\ &= \frac{E_r\left[r|f\right]}{f^2}+\frac{1-E_r\left[r|f\right]}{(1-f)^2} \\ &= \frac{1}{f}+\frac{1}{(1-f)} \text{ (since } E_r[r|f]=f) \\ &= \frac{1}{f(1-f)} \end{aligned}$$

Therefore

$$J_s = \frac{\left[\frac{\partial f(s)}{\partial s}\right]^2}{f(s)(1-f(s))}$$

(b) Assume $f(s) = \frac{1}{1+s^2}$. Plot the Fisher information on the interval $s \in [-5, 5]$. Additionally, plot the Fisher information for a Poisson neuron with the same tuning curve using the formula from the lecture. *(2 points)*

If $f(s) = \frac{1}{1+s^2}$ then

$$\left[\frac{\partial f(s)}{\partial s}\right]^2 = \left[-\frac{2s}{(1+s^2)^2}\right]^2 = \frac{4s^2}{(1+s^2)^4}$$

Thus for the Bernoulli neuron

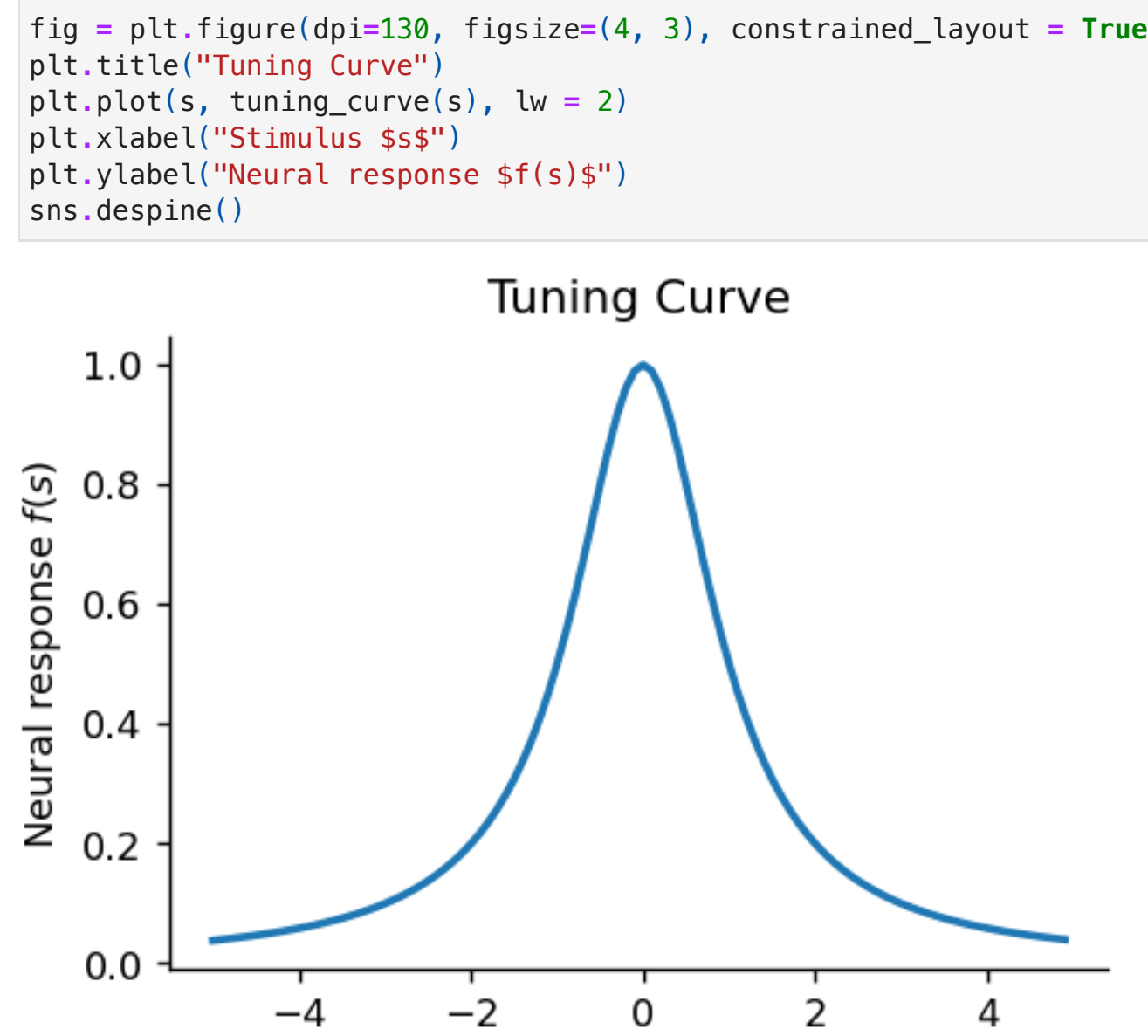
$$\begin{aligned} J_s^B &= \frac{\frac{4s^2}{(1+s^2)^4}}{\frac{1}{1+s^2}(1-\frac{1}{1+s^2})} \\ &= \frac{\frac{4s^2}{(1+s^2)^2}}{(1+s^2)-1} \\ &= \frac{4}{(1+s^2)^2} \end{aligned}$$

and for the Poisson neuron with $J_f^P = \frac{1}{f(s)}$ (c.f. lecture) $= 1+s^2$

$$J_s^P = \frac{4s^2}{(1+s^2)^4}\cdot(1+s^2) = \frac{4s^2}{(1+s^2)^3}$$

In [ ]:

```
def tuning_curve(s):
    return 1 / (1 + s**2)

s = np.arange(-5, 5, 0.1)

fig = plt.figure(dpi=130, figsize=(4, 3), constrained_layout = True)
plt.title("Tuning Curve")
plt.plot(s, tuning_curve(s), lw = 2)
plt.xlabel("Stimulus $s$")
plt.ylabel("Neural response $f(s)$")
sns.despine()
```



In [56]:

```
from scipy.special import factorial

def prob_r_bernoulli(r, s):
    return tuning_curve(s) * (r == 1) + (1 - tuning_curve(s)) * (r == 0)

def prob_r_poisson(r, s):
    return tuning_curve(s)**r / factorial(r, exact=True) * np.exp(-tuning_curve(s))
```

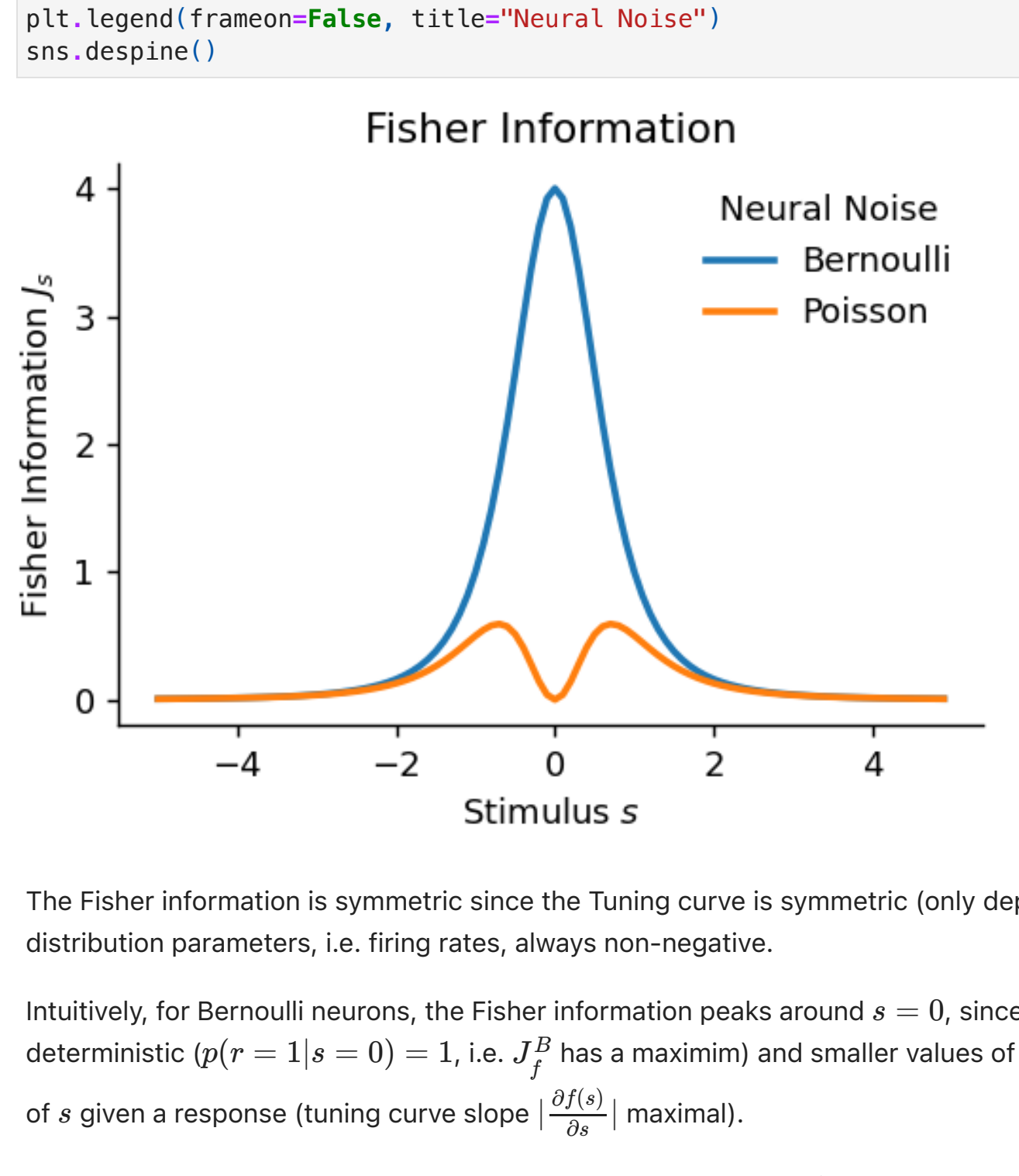
In [62]:

```
from ipywidgets import interact, FloatSlider

def plot_response_probability(s):
    fig = plt.figure(dpi=130, figsize=(4, 3), constrained_layout=True)
    plt.title("Response Probability")

    r_bernoulli = np.arange(0, 2, dtype=int)
    plt.bar(r_bernoulli, prob_r_bernoulli(r_bernoulli, s), label="Bernoulli", color="C0", widt

    r_poisson = np.arange(0, 6, dtype=int)
    plt.plot(r_poisson, prob_r_poisson(r_poisson, s), label="Poisson", color="C1", lw=2, marke

    plt.xlabel("Response $r | s$ given Stimulus $s$")
    plt.ylabel("Probability $p(r | s)$")
    plt.ylim(0, 1.1)
    plt.legend(frameon=False, title="Neural Noise")
    snb.despine()
    # plt.show()

interact(plot_response_probability, s=FloatSlider(value=1., min=0., max=5., step=0.01, descrip
plt.show()
```

```
interactive(children=(FloatSlider(value=1.0, description='s', max=5.0, step=0.01), Output()), _
dom_classes=('w…
```

In [ ]:

```
def J_tc(s):
    return 4 * s**2 / (1 + s**2)**4

def J_f_bernoulli(s):
    return 1 / (tuning_curve(s) * (1 - tuning_curve(s)))

def J_f_poisson(s):
    return 1 / tuning_curve(s)

def J_bernoulli(s):
    return 4 / (1 + s**2)**2

def J_poisson(s):
    return 4 * s**2 / (1+s**2)**3

s = np.arange(-5, 5, 0.1)

fig = plt.figure(dpi=130, figsize=(4, 3), constrained_layout = True)
plt.title("Fisher Information")
# plt.plot(s, J_tc(s), label = "Tuning Curve Factor", lw = 2)
# plt.plot(s, J_f_bernoulli(s), label = "Bernoulli Factor", lw = 2)
# plt.plot(s, J_f_poisson(s), label = "Poisson Factor", lw = 2)
plt.plot(s, J_bernoulli(s), label = "Bernoulli", lw = 2)
plt.plot(s, J_poisson(s), label = "Poisson", lw = 2)
plt.xlabel("Stimulus $s$")
plt.ylabel("Fisher Information $J_s$")
plt.legend(frameon=False, title="Neural Noise")
sns.despine()
```



The Fisher information is symmetric since the Tuning curve is symmetric (only depends on s^2) and the distribution parameters, i.e. firing rates, always non-negative.

Intuitively, for Bernoulli neurons, the Fisher information peaks around $s = 0$, since then the response becomes deterministic ($p(r=1|s=0)=1$, i.e. J_f^B has a maximim) and smaller values of $s$ allow clearer discrimination of $s$ given a response (tuning curve slope $|\frac{\partial f(s)}{\partial s}|$ maximal).

For Poisson neurons, the Fisher information peaks between $s = 1$, since for large $s$ the responses become deterministic ($p(r=0|s\to\infty)\to 1$, i.e. J_f^P is maximal) but again smaller values of $s$ allow clearer discrimination of $s$ given a response (tuning curve slope $|\frac{\partial f(s)}{\partial s}|$ maximal).

(c) Given an estimator $\hat{s}(r) = \begin{cases} 1 & if \quad r=0 \\ 0 & if \quad r=1 \end{cases}$.
Compute variance $\sigma^2(s)$, bias $b(s)$, and mean squared error $MS(s)$ of this estimator. Don't use the relationship $MS(s) = b^2(s) + \sigma^2(s)$, but instead compute them independently. *(2 points)*

Confirm for yourself with an automatized tool like wolphramalpha that their relationship is correct.

The definition implies

$$\hat{s}(r) = 1 - r \sim f(s)^{1-\hat{s}}(1-f(s))^{\hat{s}} = \text{Bernoulli}(\hat{s}|1-f(s))$$

Then

$$\begin{aligned} \sigma_{\hat{s}}^2(s) &= f(s)(1-f(s)) \\ &= \frac{1}{1+s^2}(1-\frac{1}{1+s^2}) \\ &= \frac{1}{1+s^2}\frac{s^2}{1+s^2} \\ &= \left(\frac{s}{1+s^2}\right)^2 \end{aligned}$$

Furthermore

$$\begin{aligned} b(s) &= E_{\hat{s}}\left[\hat{s}|s\right] - s \\ &= 1 - f(s) - s \\ &= \frac{s^2}{1+s^2} - s \\ &= s\left(\frac{s}{1+s^2} - 1\right) \end{aligned}$$

And

$$\begin{aligned} MS(s) &= E_{\hat{s}}\left[(s-\hat{s})^2|s\right] \\ &= \sum_{\hat{s}=0}^{1}(s-\hat{s})^2 p(\hat{s}|s) \\ &= \sum_{\hat{s}=0}^{1}(s-\hat{s})^2 f(s)^{1-\hat{s}}(1-f(s))^{\hat{s}} \\ &= s^2 f(s) + (s-1)^2(1-f(s)) \\ &= \frac{s^2}{1+s^2} + (1-s)^2\frac{s^2}{1+s^2} \\ &= \frac{s^2}{1+s^2}\left[1+(1-s)^2\right] \end{aligned}$$

Which was confirmed to satisfy the bias-variance trade-off relationship.

(d) Assume $s \sim \mathcal{U}([0,1])$. Compute the ideal observer $\hat{s}_{MS}(r)$. *(2 points)*

Hint: Compute $p(r=0)$ and $p(r=1)$ with law of total probability. Then use Bayes' theorem to compute $p(s|r)$. Useful integrals:

$$\begin{aligned} \int \frac{1}{1+s^2}ds &= \tan^{-1}(s) + c, \qquad c \in \mathbb{R} \\ \int \frac{s}{1+s^2}ds &= \frac{1}{2}\ln(1+s^2) + c, \qquad c \in \mathbb{R} \\ \int \frac{s^2}{1+s^2}ds &= s - \tan^{-1}(s) + c, \qquad c \in \mathbb{R} \\ \int \frac{s^3}{1+s^2}ds &= \frac{s^2-\ln(1+s^2)}{2} + c, \qquad c \in \mathbb{R} \end{aligned}$$

From the lecture, we know that

$$\hat{s}_{MSE}(r) := \arg\min_{\hat{s}} MSE(\hat{s}|r) = \arg\min_{\hat{s}} E_s[(\hat{s}-s)^2|r] = \arg\min_{\hat{s}} \int_0^1 (\hat{s}-s)^2\, p(s|r)\, ds$$

implies that

$$\hat{s}_{MSE}(r) = E_s[s|r] = \int_0^1 s\, p(s|r)\, ds.$$

We use Bayes Theorem to compute

$$\begin{aligned} p(s|r) &= \frac{p(r|s)\cdot p(s)}{p(r)} \\ &= \frac{p(r|f(s))\cdot 1}{\int_0^1 p(r|f(s'))ds'} \\ &= \frac{f(s)^r(1-f(s))^{1-r}}{\int_0^1 f(s')^r(1-f(s'))^{1-r}ds'} \\ &= \frac{\left(\frac{1}{1+s^2}\right)^r\left(1-\frac{1}{1+s^2}\right)^{1-r}}{\int_0^1\left(\frac{1}{1+s'^2}\right)^r\left(1-\frac{1}{1+s'^2}\right)^{1-r}ds'} \\ &= \frac{\frac{1}{(1+s^2)^r}\frac{s^{2(1-r)}}{(1+s^2)^{1-r}}}{\int_0^1\frac{1}{(1+s'^2)^r}\frac{s'^{2(1-r)}}{(1+s'^2)^{1-r}}ds'} \\ &= \frac{\frac{s^{2-2r}}{1+s^2}}{\int_0^1\frac{s'^{2-2r}}{1+s'^2}ds'} \end{aligned}$$

where the prior $p(s) = 1\ \forall s \in [0,1]$ because $s \sim \mathcal{U}([0,1])$.

Then using the given integrals

$$\begin{aligned} p(s|r=1) &= \frac{\frac{1}{1+s^2}}{\int_0^1\frac{1}{1+s'^2}ds'} \\ &= \frac{1}{(1+s^2)\cdot\tan^{-1}(s')|_{s'=0}^1} \\ &= \frac{1}{(1+s^2)\left(\tan^{-1}(1)-\tan^{-1}(0)\right)} \\ &= \frac{1}{(1+s^2)\left(\frac{\pi}{4}-0\right)} \\ &= \frac{4}{\pi(1+s^2)} \end{aligned}$$

and

$$\begin{aligned} p(s|r=0) &= \frac{\frac{s^2}{1+s^2}}{\int_0^1\frac{s'^2}{1+s'^2}ds'} \\ &= \frac{\frac{s^2}{1+s^2}}{s'-\tan^{-1}(s')|_{s'=0}^1} \\ &= \frac{\frac{s^2}{1+s^2}}{1-\frac{\pi}{4}-(0-0)} \\ &= \frac{\frac{s^2}{1+s^2}}{\frac{4-\pi}{4}} \\ &= \frac{4s^2}{(4-\pi)(1+s^2)} \end{aligned}$$

Therefore

$$\begin{aligned} \hat{s}_{MSE}(r=1) = E_s[s|r=1] &= \int_0^1 s\, p(s|r=1)\, ds \\ &= \int_0^1 s\,\frac{4}{\pi(1+s^2)}\, ds \\ &= \frac{4}{\pi}\int_0^1\frac{s}{1+s^2}\, ds \\ &= \frac{4}{\pi}\frac{1}{2}\ln(1+s^2)|_{s=0}^1 \\ &= \frac{2}{\pi}(\ln(2)-\ln(1)) \\ &= \frac{2\ln(2)}{\pi} \approx 0.44 \end{aligned}$$

and

$$\begin{aligned} \hat{s}_{MSE}(r=0) = E_s[s|r=0] &= \int_0^1 s\, p(s|r=0)\, ds \\ &= \int_0^1 s\,\frac{4s^2}{(4-\pi)(1+s^2)}\, ds \\ &= \frac{4}{(4-\pi)}\int_0^1\frac{s^3}{1+s^2}\, ds \\ &= \frac{4}{(4-\pi)}\frac{s^2-\ln(1+s^2)}{2}|_{s=0}^1 \\ &= \frac{4}{(4-\pi)}[(1-\ln(2))-(0-\ln(1))] \\ &= \frac{2(1-\ln(2))}{(4-\pi)} \approx 0.715 \end{aligned}$$

(e) Contextualize the previous exercises with Cramer-Rao bound. *(2 points)*

The Cramer-Rao bound is given by the Fisher information calculated in b) such that

$$Var\left[\hat{s}(r)|s\right] \geq \frac{g'(s)^2}{J_s^B} = \frac{(-f'(s))^2}{\frac{4}{(1+s^2)^2}} = \frac{\frac{4s^2}{(1+s^2)^4}}{\frac{4}{(1+s^2)^2}} = \left(\frac{s}{1+s^2}\right)^2$$

which is exactly the variance we calculated in part (c).

For part (d),

$$\begin{aligned} g(s) = \quad & \mathbb{E}[\hat{s}|s] \\ & \sum_{r=0}^{1}\hat{s}(r)\cdot p(r|s) \\ & \hat{s}(r=0)\cdot p(r=0|s) + \hat{s}(r=1)\cdot p(r=1|s) \\ & \hat{s}(r=0)\cdot(1-f(s)) + \hat{s}(r=1)\cdot f(s) \end{aligned}$$

therefore

$$g'(s) = \hat{s}(0)\cdot(-f'(s)) + \hat{s}(1)\cdot f'(s) = f'(s)\cdot[\hat{s}(1)-\hat{s}(0)]$$

and the biased Cramer-Rao bound is

$$\text{Var}(\hat{s}|s) \geq \frac{g'(s)^2}{J_s^B} = \frac{f'(s)^2\cdot[\hat{s}(1)-\hat{s}(0)]^2}{\frac{(f'(s))^2}{f(s)[1-f(s)]}} = [\hat{s}(1)-\hat{s}(0)]^2 f(s)[1-f(s)]$$

Now compute the variance explicitly:

$$\begin{aligned} \text{Var}(\hat{s}|s) =& \mathbb{E}[\hat{s}^2|s] - g(s)^2 \\ &= \hat{s}(0)^2\cdot(1-f(s)) + \hat{s}(1)^2\cdot f(s) - \left[\hat{s}(0)\cdot(1-f(s)) + \hat{s}(1)\cdot f(s)\right]^2 \\ &= \hat{s}(0)^2\cdot(1-f(s)) + \hat{s}(1)^2\cdot f(s) - \left[\hat{s}(0)^2(1-f(s))^2 + 2\hat{s}(0)\hat{s}(1)f(s)(1-f(s)) + \hat{s}(1)^2 f(s)^2\right] \\ &= \hat{s}(0)^2\left[(1-f(s))-(1-f(s))^2\right] + \hat{s}(1)^2\left[f(s)-f(s)^2\right] - 2\hat{s}(0)\hat{s}(1)f(s)(1-f(s)) \\ &= (\hat{s}(0)^2 + \hat{s}(1)^2 - 2\hat{s}(0)\hat{s}(1))\cdot f(s)(1-f(s)) \\ &= [\hat{s}(1)-\hat{s}(0)]^2\cdot f(s)(1-f(s)) \end{aligned}$$

Therefore, we find that

$$\text{Var}(\hat{s}|s) = [\hat{s}(1)-\hat{s}(0)]^2 f(s)(1-f(s)) \geq \frac{g'(s)^2}{J(s)} = [\hat{s}(1)-\hat{s}(0)]^2 f(s)(1-f(s))$$

and the bound is tight in this case as well.

2: Minimum Discrimination Error (MDE)

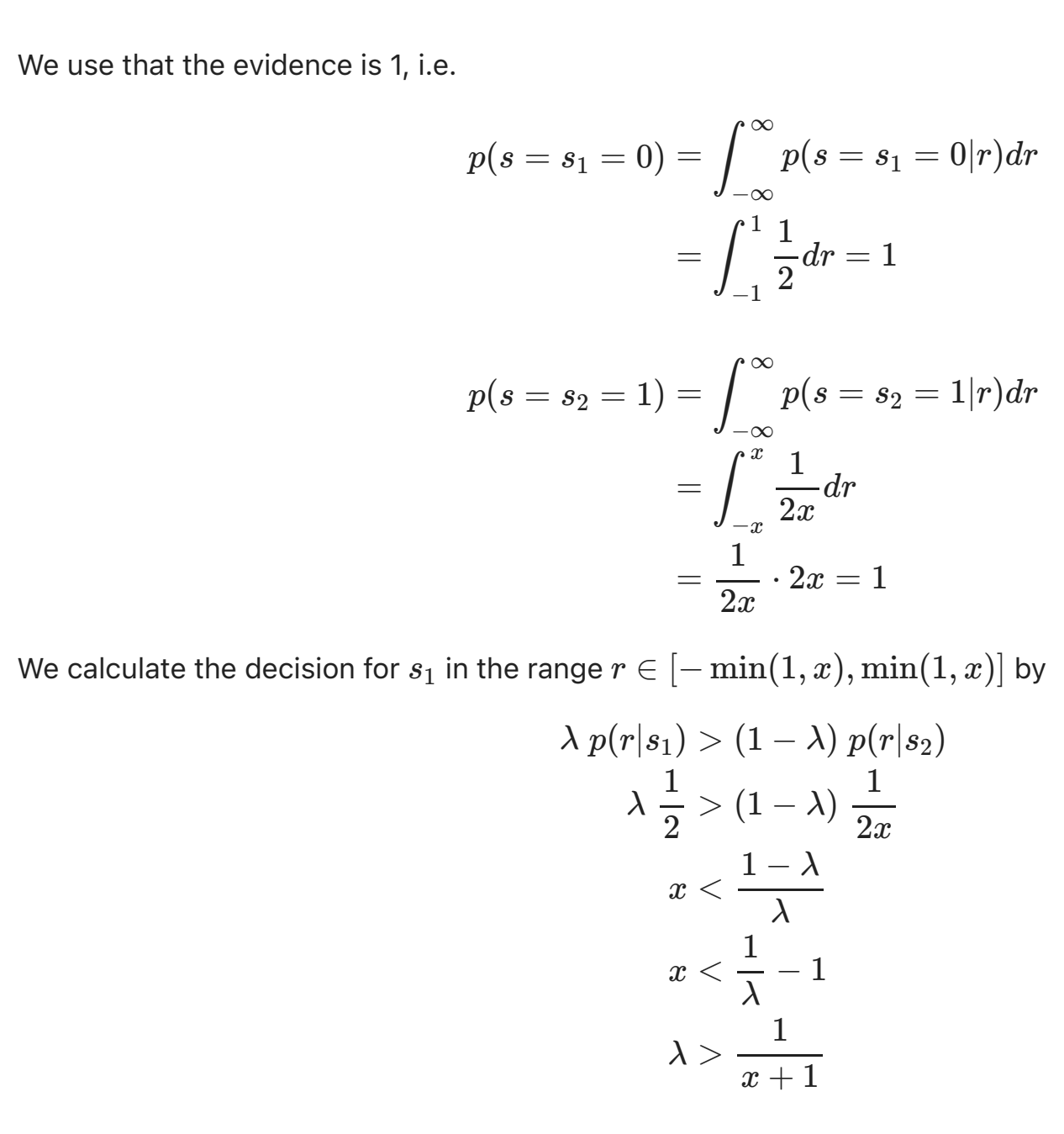
Given two stimuli s_1 and s_2 with probabilities λ and $1-\lambda$, the response function is given as

$$p(r|s) = \begin{cases} \frac{1}{2} & for \quad s=0 \quad and \quad r \in [-1,1] \\ \frac{1}{2x} & for \quad s=1 \quad and \quad r \in [-x,x] \\ 0 & else \end{cases}$$

(a) Draw diagrams for each possible case and highlight the MDE. Additionally, write down the decision for the Maximal a posteriori estimator $\hat{s}_{MAP}(r)$. *(2 points)*
Hint: Differentiate the cases by λ and x. (there are 4 possibilities)

In [170…

```
def plot_case(ax, x, lam, case_title):
    r = np.linspace(-1.5, 1.5, 500)
    # p(r|s1)
    p_s1 = np.where((r >= -1) & (r <= 1), 0.5, 0) * lam
    # p(r|s2)
    p_s2 = np.where((r >= -x) & (r <= x), 1/(2*x), 0) * (1-lam)

    ax.plot(r, p_s1, color='blue', lw=2, label='$p(s_1 | r) = \lambda \; p(r|s_1)$')
    ax.plot(r, p_s2, color='green', ls="--", lw=2, label='$p(s_2 | r) = (1-\lambda) \; p(r|s_2
    ax.fill_between(r, np.minimum(p_s1, p_s2), color='yellow', alpha=0.4, label='MDE')

    ax.set_xlim(-1.5, + 1.5)
    ax.set_ylim(0, 0.65)
    ax.set_title(case_title)

fig, axs = plt.subplots(2, 2, figsize=(7.5, 5.5), dpi=130, constrained_layout=True)
# plt.subplots_adjust(hspace=0.3, wspace=0.2)

# Case 1: x < 1, lambda < 1/(x+1)
plot_case(axs[0,0], x=0.6, lam=0.4, case_title=r'$x<1$, $\lambda < \frac{1}{x+1}$')
axs[0,0].set_ylabel('Probability $p(s|r)$')

# Case 2: x < 1, lambda > 1/(x+1)
plot_case(axs[0,1], x=0.6, lam=0.7, case_title=r'$x<1$, $\lambda > \frac{1}{x+1}$')
axs[0,1].legend(frameon=False)

# Case 3: x > 1, lambda < 1/(x+1)
plot_case(axs[1,0], x=1.3, lam=0.3, case_title=r'$x>1$, $\lambda < \frac{1}{x+1}$')
axs[1,0].set_ylabel('Probability $p(s|r)$')
axs[1,0].set_xlabel('Response $r$')

# Case 4: x > 1, lambda > 1/(x+1)
plot_case(axs[1,1], x=1.3, lam=0.6, case_title=r'$x>1$, $\lambda > \frac{1}{x+1}$')
axs[1,1].set_xlabel('Response $r$')

sns.despine()
plt.suptitle("MDE Cases", fontsize=14)
plt.show()
```



We use that the evidence is 1, i.e.

$$\begin{aligned} p(s=s_1=0) &= \int_{-\infty}^{\infty}p(s=s_1=0|r)dr \\ &= \int_{-1}^{1}\frac{1}{2}dr = 1 \end{aligned}$$

$$\begin{aligned} p(s=s_2=1) &= \int_{-\infty}^{\infty}p(s=s_2=1|r)dr \\ &= \int_{-x}^{x}\frac{1}{2x}dr \\ &= \frac{1}{2x}\cdot 2x = 1 \end{aligned}$$

We calculate the decision for s_1 in the range $r \in [-\min(1,x), \min(1,x)]$ by

$$\begin{aligned} \lambda\, p(r|s_1) &> (1-\lambda)\, p(r|s_2) \\ \lambda\,\frac{1}{2} &> (1-\lambda)\,\frac{1}{2x} \\ x &< \frac{1-\lambda}{\lambda} \\ x &< \frac{1}{\lambda}-1 \\ \lambda &> \frac{1}{x+1} \end{aligned}$$

Then we obtain for all cases (only neglecting indifferent cases) the decision

$$\begin{aligned} \hat{s}_{MAP}(r) &= \begin{cases} s_1 = 0 & \text{if } \lambda\, p(r|s_1) > (1-\lambda)\, p(r|s_2) \\ s_2 = 1 & \text{else} \end{cases} \\ &= \begin{cases} s_1 = 0 & \text{if } (x<1 \text{ and } \lambda > \frac{1}{x+1}) \\ & \text{or } (x<|r|<1 \text{ and } \lambda > \frac{1}{x+1}) \\ & \text{or } (x>1>|r| \text{ and } \lambda > \frac{1}{x+1}) \\ s_2 = 1 & \text{if } (x>1 \text{ and } \lambda < \frac{1}{x+1}) \\ & \text{or } (|r|<x<1 \text{ and } \lambda > \frac{1}{x+1}) \\ & \text{or } (x>|r|>1 \text{ and } \lambda > \frac{1}{x+1}) \end{cases} \end{aligned}$$

(b) Compute the MDE for each case analytically. *(1 point)*

For $x<1, \lambda<\frac{1}{x+1}$:

$$MDE = \int_{-x}^{x}p(s_1|r)dr = \int_{-x}^{x}\lambda\frac{1}{2}dr = x\lambda$$

For $x<1, \lambda>\frac{1}{x+1}$:

$$MDE = \int_{-x}^{x}p(s_2|r)dr = \int_{-x}^{x}(1-\lambda)\frac{1}{2x}dr = 1-\lambda$$

For $x>1, \lambda<\frac{1}{x+1}$:

$$MDE = \int_{-1}^{1}p(s_1|r)dr = \int_{-1}^{1}\lambda\frac{1}{2}dr = \lambda$$

For $x>1, \lambda>\frac{1}{x+1}$:

$$MDE = \int_{-1}^{1}p(s_2|r)dr = \int_{-1}^{1}(1-\lambda)\frac{1}{2x}dr = \frac{1-\lambda}{x}$$

(c) Confirm your results by simulation for multiple sets of parameters λ and x. Choose your selection of parameters such that they cover all cases from (a). *(2 points)*

1. Generate stimuli (sufficient amount)
2. Generate responses
3. Apply estimator (write a function of $\hat{s}_{MAP}(r)$ from (a))
4. Compute MDE
5. Compare to analytical MDE

In [ ]:

```
def generate_stimuli(N, lam):
    # s1=0 with prob lam, s2=1 with prob 1-lam
    return np.random.choice([0, 1], size=N, p=[lam, 1-lam])

def generate_responses(stimuli, x):
    # For s=0: r ~ Uniform[-1, 1]
    # For s=1: r ~ Uniform[-x, x]
    r = np.empty_like(stimuli, dtype=float)
    mask0 = (stimuli == 0)
    mask1 = (stimuli == 1)
    r[mask0] = np.random.uniform(-1, 1, size=mask0.sum())
    r[mask1] = np.random.uniform(-x, x, size=mask1.sum())
    return r

def s_map_estimator(r, lam, x):
    # Decision rule from a)
    # For r in [-min(1, x), min(1, x)], compare lambda/2 vs (1-lambda)/(2x)
    # Otherwise, assign to the class whose support contains r
    s_hat = np.zeros_like(r)
    min1x = min(1, x)
    # In overlap region
    overlap = (r >= -min1x) & (r <= min1x)
    # Compute threshold
    if lam > 1/(x+1):
        s_hat[overlap] = 0
    else:
        s_hat[overlap] = 1
    # Outside overlap, assign by support
    s_hat[(r < -x) | (r > x)] = 0  # only s=0 possible
    s_hat[(r < -1) | (r > 1)] = 1  # only s=1 possible
    return s_hat

def analytical_mde(lam, x):
    if x < 1 and lam < 1/(x+1):
        return x * lam
    elif x < 1 and lam > 1/(x+1):
        return 1 - lam
    elif x > 1 and lam < 1/(x+1):
        return lam
    elif x > 1 and lam > 1/(x+1):
        return (1 - lam) / x
    else:
        return np.nan

# Parameters for all 4 cases
params = [
    # (x, lam, description)
    (0.6, 0.4, "Case 1: $x<1$, $λ<1/(x+1)$"),
    (0.6, 0.7, "Case 2: $x<1$, $λ>1/(x+1)$"),
    (1.3, 0.3, "Case 3: $x>1$, $λ<1/(x+1)$"),
    (1.3, 0.6, "Case 4: $x>1$, $λ>1/(x+1)$"),
]

N = 100000
np.random.seed(42)
for x, lam, desc in params:
    stimuli = generate_stimuli(N, lam)
    responses = generate_responses(stimuli, x)
    s_hat = s_map_estimator(responses, lam, x)
    mde_sim = np.mean(s_hat != stimuli)
    mde_analytical = analytical_mde(lam, x)
    print(f"{desc}: Simulated MDE = {mde_sim:.4f}, Analytical MDE = {mde_analytical:.4f}")
```

```
Case 1: $x<1$, $λ<1/(x+1)$: Simulated MDE = 0.2382, Analytical MDE = 0.2400
Case 2: $x<1$, $λ>1/(x+1)$: Simulated MDE = 0.2999, Analytical MDE = 0.3000
Case 3: $x>1$, $λ<1/(x+1)$: Simulated MDE = 0.2987, Analytical MDE = 0.3000
Case 4: $x>1$, $λ>1/(x+1)$: Simulated MDE = 0.3085, Analytical MDE = 0.3077
```

We validate the analytic solutions by observing closeby results of the simulation.